

aLOTofImaginArches: interactive graphical statics of masonry arches in the browser

G. Castellazzi

Department of Civil, Chemical, Environmental and Materials Engineering (DICAM), University of Bologna, Viale Risorgimento 2, 40136 Bologna, Italy

`giovanni.castellazzi@unibo.it`

Abstract

The equilibrium of a masonry arch is not a single state but a three-parameter family of them, and that multiplicity is the hardest idea to convey when the subject is taught. *aLOTofImaginArches* makes it manipulable. A student loads a photograph of a real arch, traces its intrados and extrados, subdivides the ring into voussoirs, sets a scale, applies loads, and then moves the three degrees of freedom of the line of thrust while the software reports, joint by joint, whether the line stays within the masonry: Heyman's safe theorem, made operable. The software is a rewrite of an established MATLAB application as dependency-free ES modules that run directly from a static host, so that no licence and no installation stand between a student and the construction. The graphical idiom of the original is preserved deliberately. Twenty-seven legacy examples are carried over and fifteen of them, which store a complete solution, are reproduced to machine precision.

Keywords: masonry arches, line of thrust, graphical statics, limit analysis, engineering education, web application

1. Motivation and significance

A masonry arch stands not because it is in *the* state of equilibrium but because it is in *one* of infinitely many. Under Heyman's assumptions — no tensile strength, unlimited compressive strength, no sliding — the safe theorem states that if any line of thrust in equilibrium with the applied loads can be found lying wholly within the masonry, the structure is safe [1, 2, 3]. For a two-dimensional arch that line has three degrees of freedom, the ∞^3 of the classical literature [4, 5].

Students meet this in a lecture and, in the author's experience of teaching it, do not believe it. The obstacle is not the algebra of the funicular polygon, which is elementary, but the multiplicity: a single worked example produces a single drawing, and a single drawing looks like a single answer. What convinces is moving the solution and watching it remain a solution — and then watching it stop being an *admissible* one as it leaves the ring.

Interactive equilibrium tools have a distinguished record in exactly this role [6, 7, 8], and this software follows that line for the two-dimensional arch, with an emphasis on the historical construction rather than on generality. Its predecessor, a MATLAB App used

for several years in the master's courses *Mechanics of Historical Masonry Structures* and *Historical Masonry and Wooden Structures* at Bologna, was limited by two things: it required a MATLAB licence, which excludes students working on their own machines and anyone outside a subscribing institution, and its source, packaged in a binary `.mlapp` container, was not something a student could read.

This paper describes a rewrite as a browser application: plain ES modules with no dependencies and no build step, served as static files, so that the source a student reads is exactly the source that runs. Three commitments shaped it.

1. **The graphics of the original are preserved deliberately.** The boxed axes with inward ticks, the pale grid, the translucent force bands and the constant-size arrowheads of the MATLAB figure are reproduced on the HTML canvas. Continuity of appearance is not nostalgia: the course notes, the slides and the existing tutorials all show that idiom, and a student switching between them should not have to re-learn what a picture means.
2. **The mechanics is separated from the drawing and is tested.** The construction lives in seven small

Table 1: Code metadata

Nr.	Code metadata description	Please fill in this column
C1	Current code version	v1.0.0
C2	Permanent link to code/repository used for this code version	https://github.com/gcastellazzi/aLOTofImaginArches
C3	Permanent link to Reproducible Capsule	The application itself is the capsule: it is served as static files and runs, unmodified, at https://gcastellazzi.github.io/aLOTofImaginArches/app/
C4	Legal Code License	MIT License
C5	Code versioning system used	git
C6	Software code languages, tools, and services used	JavaScript (ES2022 modules), HTML5 Canvas, CSS; Python 3 for the legacy-format converter; GitHub Pages
C7	Compilation requirements, operating environments, dependencies	None. Any modern browser runs the application as served; there is no build step and no third-party runtime dependency. Node.js ≥ 18 is needed only to run the test suite; Python 3 with NumPy and SciPy only to re-run the converter.
C8	Link to developer documentation/manual	https://gcastellazzi.github.io/aLOTofImaginArches/ and the README.md files under docs/app/
C9	Support email for questions	giovanni.castellazzi@unibo.it

modules with no reference to the DOM, so it can be, and is, exercised by a test suite from the command line.

3. **The legacy examples are carried over, and checked.** Every stored example of the MATLAB version was converted, and the conversion was verified field by field rather than assumed.

2. Software description

2.1. Software architecture

The application is a static web page. There is no server, no bundler and no package to install: the browser loads `index.html`, which imports ES modules directly.

The mechanics lives in nine core modules – geometry, blocks, statics, tracing, units, the saved-model reader, the save format, the mechanism and the dome – with three more for the drawing and one for the interface: about 4 500 lines in all.

The separation is strict in one direction: `core` knows nothing of the canvas, and `render` knows nothing of arches. It is what allows the mechanics to be checked without a browser. The suite comprises 123 tests run by the Node test runner (`node -test`), covering the geometry, the funicular construction, the reproduction of stored examples, tracing, scale and units, admissibility, hinge formation and the collapse kinematics, the dome, and the save format.

A separate Python tool, `tools/mat2json.py`, converts the MATLAB `.mat` examples to JSON and extracts the background images. It carries a `-check` mode that re-reads every converted file and compares it field

by field with the original, so the conversion is verified rather than trusted.

2.2. Software functionalities

Stored examples. Twenty-eight example files are shipped. One is an empty template; of the remaining twenty-seven, fifteen carry a complete stored solution and are recomputed from their geometry alone, six were saved before a solution was computed, and six are internally inconsistent — the stored solution does not correspond to the stored geometry. The software detects the last group and says so, rather than drawing a solution it cannot justify (Section 3.1).

Tracing. A student loads any image and clicks along the intrados and the extrados. The two curves are re-sampled by arc length and joined into n voussoirs with $n + 1$ radial joints. The trace is checked before it is used: too few points, coincident curves, and — the case that matters — curves that cross are all reported. A crossing is detected by the *sign* of the signed area of the resulting blocks, not by their size, because an inverted block can be as large as a sound one.

Scale, units and loads. Until a scale is set, every quantity is in pixels. The student picks two points and states the real distance between them; lengths then scale by k , areas by k^2 , and weights by k^2 or k^3 according to whether the out-of-plane thickness was given physically or in pixels — a distinction worth making explicit, since getting it wrong is a silent factor of the scale. Three unit systems are offered, matching the original. Span and rise are reported after scaling, being the quickest

check that the reference distance was entered correctly. Point loads are placed by clicking; a load and a voussoir weight are stations on the load line alike, so loads are merged into the same ordered sequence as zero-area blocks.

The three degrees of freedom. The pole of the force polygon has two coordinates – its abscissa is the horizontal thrust, its ordinate divides the total weight between the two vertical reactions – and the line has one free end, sliding along its springing joint from the intrados to the extrados. Each is a slider. The far end is not a fourth parameter: it follows, because the last ray is carried on until it meets the other joint, and where it lands is reported.

Admissibility. For each joint the software computes where the line crosses it, as a fraction s of the joint. The verdict names the number of joints violated, the worst one and on which face, or — when the line fits — how much room is left. Choosing the crossing correctly is less trivial than it appears: a joint near a springing is almost horizontal, and its infinite line meets segments of the thrust line far away that have nothing to do with it, so the implementation prefers the crossing that lies *within* the joint.

Mechanism. The line cannot leave the ring: where it would, it stays hooked at the intrados or the extrados between two voussoirs, and that point is a hinge. Counting the two springings as hinges throughout, h hinges carry $h - 1$ rigid macro-blocks and the arch has $3(h - 1) - 2h = h - 3$ degrees of freedom: two hinges is the once-hyperstatic arch whose equilibrium state is not determined, three is the three-pin arch, four is a mechanism. In this mode the thrust slider alone commands the line, the other two parameters being chosen to hold it as far from both faces as it will go, so that hinges appear only when the thrust forces them. The collapse kinematics is drawn (Section 3.3).

Poleni's dome. A barrel vault cut into voussoirs gives blocks of constant width. A dome cut into lunes does not: each lune is bounded by two meridian planes, so its width is proportional to the distance from the axis, broad at the major parallel and closing to nothing at the crown. With the option set, the weight of each voussoir is taken from Pappus' theorem, $V = A\theta\bar{r}$ with $\bar{r}$ the distance of the block's centroid from the axis — exact for a plane region turned about an axis in its plane. The axis defaults to the mid-point of the springings and can

be picked on the drawing. A second view of the right-hand pane shows the voussoirs as solids, prismatic or revolved.

Hooke's cable. The same polygon reflected about the chord joining the two springings, so that the cable hangs from the points the arch springs from whatever their relative height, and *ut pendet continuum flexile, sic stabit contiguum rigidum inversum* [9] can be shown rather than quoted. The rays of the force polygon are drawn dashed and lettered after Bow, with the same letter on the parallel segment of the thrust line, so that a force can be carried from one drawing to the other by eye.

Saving. The whole session — traced curves, blocks, joints, weights, loads, scale, unit system, dome settings and the three parameters — is written to a single JSON file and read back. Everything derivable is deliberately omitted and recomputed on opening, so a file stays readable when the mechanics is corrected; a file is validated before it is trusted and refused with a diagnostic rather than half-loaded.

3. Illustrative examples

3.1. Reproducing Poleni

Figure 1 shows the software working over Poleni's own plate of 1748 for the dome of St Peter's [10], the drawing in which the hanging-chain analogy was first put to a real structure. The 56 voussoirs and their weights come from the stored example; the force polygon and the line of thrust are recomputed from that geometry by the JavaScript implementation. The largest discrepancy between the recomputed line and the one MATLAB saved is 8.0×10^{-16} relative — machine precision. Across all fifteen examples that carry a complete solution, the worst relative discrepancy is 8.5×10^{-16} for the force polygon and 3.9×10^{-15} for the line of thrust.

That agreement is worth stating precisely because the remaining examples do *not* agree, and for identifiable reasons: in some the stored force polygon was drawn with weights that no longer match the stored blocks; in others, applied loads were merged into the sequence but the loads themselves were never written to the file. This is a property of the original save format, not of either implementation, and it was found only because the port was checked against the original rather than eyeballed. The software now reports these files as not recomputable and displays the stored solution with a warning.

Table 2: Least t/r_i admitting a line of thrust on a semicircular ring, by bisection, against the number of voussoirs n . Heyman’s value for a continuous ring is 0.108.

n	ends pinned	ends free
8	0.1789	0.1116
12	0.1894	0.1118
16	0.1984	0.1155
24	0.2070	0.1181
32	0.2061	0.1179
48	0.2085	0.1199
64	0.2093	0.1202

The dome Poleni was actually analysing

Reproducing the stored file exercises the port; it does not exercise the mechanics Poleni was after. The plate is a section of a *dome*, and a lune of a dome is not a slice of a barrel: its width grows with the distance from the axis. Setting the Poleni option on the same 56 voussoirs, with the axis where the software places it by default, moves **14.7 per cent of the total weight** from the upper courses to the lower ones. The outermost block rises from 4.15 to 5.56 per cent of the total and the innermost falls from 1.29 to 0.34. The line of thrust follows, and the two idealisations of the same drawing part company — which is the whole of Poleni’s argument, and something the constant-thickness reading cannot show.

3.2. The minimum thickness of a semicircular arch

The second example is the classical one, and it exposed a defect worth recording. For a semicircular ring of uniform thickness the least t/r_i at which an admissible line exists is about 0.108 [3, 11].

An early version of this software pinned both ends of the line at the mid-points of the end joints, leaving only the thrust free. Measured by bisection on a ring of 16 voussoirs, that one-parameter family first admits a line at $t/r_i = 0.198$ — nearly twice Heyman’s value. The tool was therefore condemning arches that stand.

Two things were wrong. The ends were pinned, discarding two degrees of freedom; and, less obviously, the funicular construction inherited from the original stops on the *vertical* through the far springing rather than reaching the far joint at all, so asking where the line crossed that joint returned the intersection of the joint’s infinite line with an unrelated segment. Freeing the start along its joint and carrying the last ray on to the far joint restores the full family. Table 2 gives the measured result.

With the ends free the figure settles at about 0.12 and approaches Heyman’s 0.108 from above; the residual gap is discretisation, a funicular polygon of n segments

being able to cut corners that a continuous curve cannot, and it narrows as n falls rather than rises for exactly that reason. More telling than the number is that the search, given no guidance, recovers the textbook limit state on its own: the critical line leaves one springing at the *extrados*, arrives at the other at the *extrados*, and requires a horizontal thrust of about one fifth of the total weight.

Figure 2 gives the admissible range of thrust against thickness for both families: the safe theorem, drawn. The free band opens at $t/r_i \approx 0.12$ and has already widened to $0.190 \leq H/W \leq 0.215$ at 0.14, a thickness at which the pinned family still admits nothing at all; the pinned band does not open until 0.20.

3.3. Hinges, and what a degree of freedom does not tell you

Driven from the thrust, the same ring produces the classical collapse patterns without being told what to look for. At $t/r_i = 0.18$ the arch stands for $0.171 \leq H/W \leq 0.238$. At the lower limit five hinges have formed — the two springings, the intrados at the haunches and the extrados at the crown — giving four macro-blocks and two degrees of freedom; at the upper limit, four hinges and one. In between, the line runs clear of both faces, nothing is located, and the arch is once hyperstatic: its equilibrium state is not determined, which is the ∞^3 restated in the language of constraints.

Figure 3 shows the minimum-thrust mechanism displaced. Two things about that drawing were harder than they look.

The first is that an instantaneous centre is instantaneous. Turning each macro-block about its centre through a *finite* angle keeps the blocks rigid but opens the hinges at second order — on a chain of span 8, an amplitude of 0.2 radians parts them by 0.14, which is plainly visible. The motion is therefore integrated: at every step the velocity field is re-solved for the current hinge positions and each block advanced a little about its current centre, so every block undergoes a composition of rigid transforms and stays exactly rigid while the residual opening falls as amplitude²/steps, to 1.2×10^{-3} at the default.

The second is that **masonry cannot pass through itself**, and the sign of a null-space vector falls out of the elimination rather than out of the mechanics. A hinge sits at one face of its joint and the joint must open at the other, so the sense of the motion is taken from the joints, not from the arithmetic. Some hinge patterns admit *no* good sense: on a symmetric ring at maximum thrust both haunch hinges fall on the intrados, which leaves the crown block turning about a point on the axis instead of dropping, and one haunch opens exactly as the

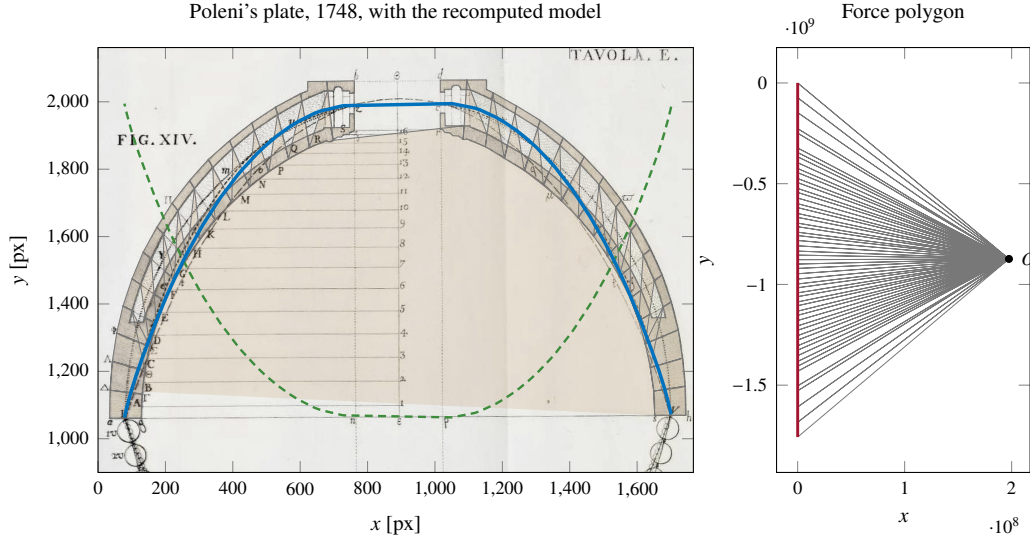


Figure 1: Poleni's plate of 1748 with the model recomputed by the software: 56 voussoirs (grey), the line of thrust (blue) and the inverted Hooke cable (green), with the force polygon on the right. The line is recomputed from the geometry alone and agrees with the solution stored by the MATLAB application to 8.0×10^{-16} relative.

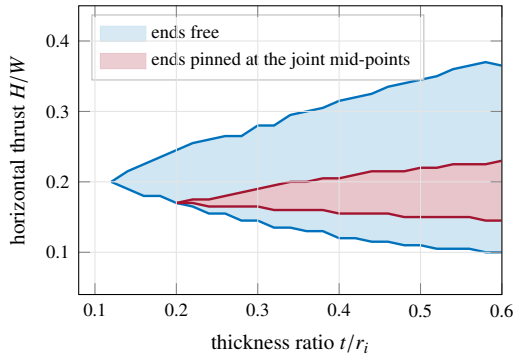


Figure 2: Range of horizontal thrust, as a fraction of the total weight, for which an admissible line exists, against the thickness ratio. Freeing the ends both lowers the thickness at which the band opens and widens it everywhere.

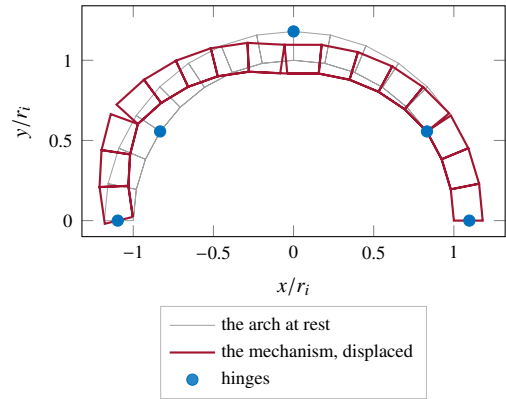


Figure 3: The minimum-thrust collapse mechanism of a semicircular ring, $t/r_i = 0.18$, $n = 16$, at $H/W = 0.171$: 5 hinges, 4 rigid macro-blocks, 2 degrees of freedom. The displacement is integrated rather than applied as a single rotation, so the blocks stay rigid and every joint opens instead of interpenetrating.

other shuts. The software reports that the pattern would need the masonry to interpenetrate rather than animating it. *A positive degree of freedom is not by itself a collapse mode*: the joints have to be able to open, and the count alone does not say whether they can.

4. Impact

The immediate impact is on teaching. The software is used in the master's course *Mechanics of Historical Masonry Structures* at the University of Bologna, alongside

a set of lecture slides, handouts and exercises derived from [12]. Removing the licence requirement changes what can be asked of students: an exercise can now be set as *photograph an arch in this city, trace it, and find an admissible line*, and every student can do it on their own machine, at home, without an installation. The saved session is a single small JSON file, which makes such an exercise collectable and markable.

A second effect is on reproducibility. Because the

port was verified against the original rather than reimplemented from its description, it produced a usable audit of a decade of stored examples and identified six that cannot be reproduced from what they contain — a researcher relying on them would have had no warning. That a rewrite is an opportunity to audit the data of the original, and that the audit should be automated and kept, applies well beyond this software. Third, the code is small enough to be read: a 300-line module in which `forcePolygon` and `funicular` sit side by side with the classical terms as their variable names is itself a teaching artefact.

The limitations follow from Heyman’s assumptions rather than from the implementation: the analysis is two-dimensional, blocks are rigid, sliding is not checked, and the finite subdivision makes the computed line a polygon inscribed in the true funicular curve. None are hidden from the user; the discretisation, in particular, is a slider.

5. Conclusions

aLOToImaginArches makes the three-parameter family of equilibrium states of a masonry arch something a student can take hold of and move. The rewrite from MATLAB to dependency-free browser modules removes the licence and the installation, preserves the graphical language of the original, and — by separating the mechanics from the drawing — makes the construction testable from the command line. The legacy examples are reproduced to machine precision where they can be reproduced at all, and the software says so when they cannot.

Freeing the two end positions of the line, which the original construction had fixed, brought the computed minimum thickness of a semicircular ring from $t/r_i = 0.198$ to 0.115 against Heyman’s 0.108 , and let the software recover the classical limit state without being told what to look for. The same holds of the collapse patterns, which the hinge analysis finds unaided, and of Poleni’s own question: treating his plate as the dome it is, rather than as a barrel of the same profile, redistributes a seventh of the weight. Work in progress extends the treatment to arches on spreading supports and to the sliding condition that Heyman’s third assumption sets aside.

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